

Vishal Chand

vishalchandcv@gmail.com | +353-872673797

EDUCATION

IIT ROORKEE

INTEGRATED DUAL DEGREE (B.TECH
+ M.TECH)

2010 - 2015 | Roorkee, India

Cum. GPA: 8.677 / 10.0

LINKS

Personal:// vishalchand.com

Github:// [vish-chan](https://github.com/vish-chan)

LinkedIn:// [vishal-chand](https://www.linkedin.com/in/vishal-chand)

SKILLS

JVM & RUNTIME

OpenJDK • JVM Internals • Hotspot VM
GC Design • Module Systems • Bytecode
(ASM)

PERFORMANCE ENGINEERING

Benchmarking (JMH, SpecJBB)
Profiling (JFR, async-profiler, perf)
Performance Modeling • Bottleneck
Analysis

CLOUD & INFRASTRUCTURE

AWS (EC2, S3, EKS) • Cost Optimization
Distributed Systems • Kubernetes
Load Testing • Capacity Planning

LANGUAGES

Expert: Java • C++

Proficient: Python • C • Go

Exposure: Rust • JavaScript • Android

TOOLS & FRAMEWORKS

Perf • Systrace • GDB • Git
AMD uProf • Scikit-learn

EXPERIENCE

AMAZON | SOFTWARE DEVELOPMENT ENGINEER II

Languages and Runtime | 2022 – Present | Dublin, Ireland

- **Performance JVM Internals:** Optimized `CustomModuleLoader` reducing latency by **96.8% (120 μ s to 3 μ s)**, unblocking Athena's JDK 11.17 adoption; resolved Reflection API bottlenecks for a **40% improvement** in Jython benchmarks.
- **Distributed Systems Scaling:** Validated Pyroscope V2 horizontal scalability (1,700 profiles/s) on a 74-node cluster; isolated Raft consensus constraints and re-architected query-backend concurrency for linear throughput scaling.
- **FinOps & Cost Strategy:** Identified **20–36% savings** via instance optimization and S3 Intelligent-Tiering; modeled cost-per-TPS efficiency across 30–365 day retention periods.
- **Benchmarking Automation:** Built end-to-end benchmarking automation on AWS EC2 with automated provisioning and analysis, reducing manual benchmarking toil by **80%**.
- **Systems Innovation:** Prototyped a **zero-overhead code coverage tool** leveraging JVM profiling data, eliminating instrumentation overhead and enabling production-safe agentless coverage.
- **JDK Migration Acceleration:** Designed an automated bytecode/API analyzer for JDK migrations, reducing development cycles by **35%**.
- **Mentorship:** Guided two interns through high-complexity projects in the Languages & Runtime org, resulting in one receiving a full-time offer and the deployment of JVM flags translation feature.

AMD | SENIOR SOFTWARE SYSTEM DESIGNER

Java Runtime & Performance | 2020 – 2022 | Bengaluru, India

- **OpenJDK Core Contribution (JDK-8272773):** Architected and implemented the **Configurable Card Size** feature for the G1 Garbage Collector in OpenJDK-18; optimized evacuation pause times for ultra-large heap server workloads and improved SpecJBB2015 score by 2%.
- **Performance Modeling:** Led characterization and tuning of industry-standard enterprise benchmarks (**SPECjbb2015**, **SPECpower**, **Renaissance**) for the EPYC processor roadmap, identifying critical Hotspot VM optimization opportunities.
- **AutoTuner Framework:** Designed AutoTuner, an intelligent benchmark optimization framework using black-box algorithms to tune benchmark configuration and JVM flags, increasing tuning precision and reducing manual effort from **14 days to 3 days**.
- **Community Impact:** Contributed to Amazon Corretto's **Heapothesys** benchmark suite.
- **Mentorship:** Led a postgraduate intern in architecting the AutoTuner Feature Selection module; reduced tuner search-time by 50% while maintaining 99%+ optimization accuracy.

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ACHIEVEMENTS

- 2022 DSG Excellence (AMD)
- 2021 Spotlight (AMD)
- 2013-17 Qualstar (Qualcomm)
- 2014 Winner, IdeaQuest

GRADUATE

COURSEWORK

- Advanced Algorithms
- Advanced Operating Systems
- Advanced Computer Architecture
- Advanced DBMS
- Machine Learning (Stanford)

TECHNICAL EXPERTISE

- **Runtime Systems:** OpenJDK, Hot-spot VM, GC algorithms
- **Performance:** Profiling, benchmarking, optimization
- **Distributed Systems:** Scalability, Raft consensus
- **Cloud Infrastructure:** AWS, cost optimization, FinOps
- **Observability:** async-profiler, JFR, Grafana Pyroscope

EXPERIENCE (CONTD.)

EXPERITEST | SOFTWARE ENGINEER

Seetest Android | 2018 – 2019 | Delhi, India

- Engineered core subsystems for Seetest Manual, a leading mobile automation platform.
- Conducted high-level POCs and competitive analysis for cloud-based mobile testing infrastructure.

QUALCOMM | SOFTWARE ENGINEER

Android Performance | 2015 – 2018 | Hyderabad, India

- Spearheaded software performance tasks for Qualcomm's Wearable SOC product line (Snapdragon Wear).
- Developed a **Boot Analyzer** in Python for USE analysis, driving a 10% reduction in Android boot-up time via parallelized package scanning.
- Designed an **ML-based CPU Workload Classifier** using Scikit-learn to dynamically optimize power-performance states.
- Investigated Android rendering pipelines using Systrace to solve frame-drop issues in Wear OS.
- Implemented "Suspend to Disk" (Hibernation) feature POC for power-constrained wearable devices.

NOTABLE OPEN SOURCE CONTRIBUTIONS

OPENJDK | CONTRIBUTOR

- **JDK-8272773:** Implemented configurable card table sizing (128B-1024B) for G1, Parallel, and Serial GCs via **GCCardSizeInBytes** flag, optimizing Remembered Set overhead for large heaps.
- **JDK-8307732:** Backported build-test-lib infrastructure to JDK 17u, enabling WhiteBox testing library for jtreg test suite execution.

PERFORMANCE TOOLING | COMMUNITY CONTRIBUTOR

- **async-profiler:** Implemented PMU counter scaling for kernel time multiplexing, improving concurrent profiler measurement accuracy.
- **Heapothesis:** Fixed survivor queue ordering and integer overflow for allocations $\geq 1024\text{MB}$ in Amazon Corretto's GC benchmark tool.